

Math 121: Commutative Algebra

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Worksheet 1.2: Quotients, Primes, Maximal Ideals

Problem 1. Operations on Ideals.

Throughout this problem let I and J be ideals in a ring R .

- (a) Prove or disprove that $I \cap J$ is an ideal.
- (b) Prove or disprove that $I \cup J$ is an ideal.
- (c) Prove or disprove that the product of ideals, defined below, is an ideal:

$$IJ := \{a_1b_1 + \cdots + a_tb_t \mid a_i \in I, b_i \in J, \text{ for all } t\}.$$

- (d) Prove or disprove that the sum of ideals, defined below, is an ideal:

$$I + J := \{a + b \mid a \in I, b \in J\}.$$

- (e) Prove or disprove that the radical of an ideal, defined below, is an ideal:

$$\sqrt{I} := \{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{Z}_{\geq 1}\}.$$

Solution.

- (a) $I \cap J$ is an ideal.

Let $a, b \in I \cap J$, $r \in R$.

- (i) $I \cap J$ is a subgroup because intersection of subgroups is a subgroup.
- (ii) Note that $a, b \in I, J$. Since I and J are closed under addition and inverses, $a - b \in I, J$. Hence $a - b \in I \cap J$.
- (iii) $r \cdot a \in I \cap J$ since $ra \in I$ and $ra \in J$.

Hence $I \cap J$ is an additive subgroup of R and is closed under multiplication from elements of R .

(b) $I \cup J$ is not an ideal.

Consider the ideals $I = \langle 2 \rangle$, $J = \langle 3 \rangle$ in the ring $R = \mathbb{Z}$. Note that $5 \notin I \cup J$ even though $2, 3 \in I \cup J$. Hence, $I \cup J$ is not closed under addition and is not an ideal.

(c) IJ is an ideal.

Let $s \in IJ$, $r \in R$.

(i) By definition, IJ includes all finite sums of elements in I, J . So it contains 0 and is closed under addition and inverses. Hence it is an additive subgroup.

(ii) Consider $r \cdot s \in IJ$.

$$r \cdot s = r(a_1 b_1 + \cdots + a_t b_t) = r \cdot a_1 b_1 + \cdots + r \cdot a_t b_t$$

Note that $r \cdot a_i \in I$ for all i and thus $rs \in IJ$ for all r .

Hence IJ is an additive subgroup of R and is closed under multiplication from elements of R .

(d) $I + J$ is an ideal.

Let $p = a_1 + b_1$, $q = a_2 + b_2 \in I + J$, $r \in R$.

(i) Note that $0 \in I + J$ since $0 \in I, J$.

(ii) Consider $p - q$.

$$p - q = a_1 + b_1 - (a_2 + b_2) = (a_1 - a_2) + (b_1 - b_2)$$

Note that $(a_1 - a_2) = a$ for some $a \in I$ and $(b_1 - b_2) = b$ for some $b \in I$. Hence, $I + J$ contains zero and is closed under addition and inverses and thus is an additive subgroup.

(iii) Consider $r \cdot p$.

$$r \cdot p = r \cdot (a_1 + b_1) = ra_1 + rb_1$$

Note that $ra_1 \in I$ and $rb_1 \in J$ by definition of ideals. Hence $rp \in I + J$.

Thus $I + J$ is an additive subgroup of R and is closed under multiplication from elements of R .

(e) $\sqrt{I}$ is an ideal.

Let $a^n, b^m \in I$, and $a, b, r \in R$.

(i) Note that $0 \in \sqrt{I}$ since $0^1 \in I$.

(ii) Consider $(a - b)^{nm}$.

$$(a - b)^{nm} = \sum_{i=0}^{nm} \binom{nm}{i} (a)^i (-b)^{nm-i}$$

- If $i \geq n$, then $a^n \in I \implies \binom{nm}{i} (a)^i (-b)^{nm-i} \in I$.
- If $i < n$, then $nm - i > nm - n = n(m - 1) = nm - i \geq m$. And $b^m \in I \implies \binom{nm}{i} (a)^i (-b)^{nm-i} \in I$.

Since every term in the sum lies in I , the whole sum lies in I . So $a - b \in \sqrt{I}$.

(iii) Note that if $a^n \in I$ then $(ra)^n \in I$. Hence $ra \in \sqrt{I}$.

Thus $\sqrt{I}$ is an additive subgroup of R and is closed under multiplication from elements of R .

Problem 2. Let $\mathbb{K}$ be a field. The rings $\mathbb{Z}$, $\mathbb{K}[x]$, and $\mathbb{K}[[t]]$ are all principal ideal domains (PIDs) meaning domains where every ideal is generated by a single element. If $I = \langle a \rangle$ and $J = \langle b \rangle$ give, with proof, a careful concrete description of generators for IJ , $I \cap J$, $I + J$, and $\sqrt{I}$ in each of these rings.

Solution.

Let $I = \langle a \rangle$, $J = \langle b \rangle$. Then, we have

(a) $\mathbb{Z}$

- (i) $IJ = \langle ab \rangle$
- (ii) $I \cap J = \langle \text{lcm}(a, b) \rangle$
- (iii) $I + J = \langle \text{gcd}(a, b) \rangle$
- (iv) Let $a = p_1^{e_1} \cdots p_k^{e_k}$, then $\sqrt{I} = \langle p_1 \cdots p_k \rangle$, where p_i is a prime element.

(b) $\mathbb{K}[x]$

- (i) $IJ = \langle ab \rangle$
- (ii) $I \cap J = \langle \text{lcm}(a, b) \rangle$
- (iii) $I + J = \langle \text{gcd}(a, b) \rangle$
- (iv) Let $a = p_1^{e_1} \cdots p_k^{e_k}$, then $\sqrt{I} = \langle p_1 \cdots p_k \rangle$, where p_i is an irreducible element.

(c) $\mathbb{K}[[t]]$ Note that elements of $\mathbb{K}[[t]]$ are of two forms: $a \in \mathbb{K}$ or $a \cdot t^n$, $a \in \mathbb{K}$. So all proper ideals are of the form $\langle 0 \rangle$ or $\langle t^n \rangle$. Let $I = \langle t^n \rangle$, $J = \langle t^m \rangle$. If I is trivial, then $IJ = I \cap J = \sqrt{I} = \langle 0 \rangle$, and $I + J = J$. Assuming neither I nor J are trivial we have,

- (i) $IJ = \langle t^{n+m} \rangle$
- (ii) $I \cap J = \langle t^{\max(n, m)} \rangle$
- (iii) $I + J = \langle t^{\min(n, m)} \rangle$
- (iv) $\sqrt{I} = \langle t \rangle$.

Universal Properties. Throughout this course we will see numerous constructions and operations (e.g., quotients, tensor products, localizations, completions, etc.) each of these will have an explicit construction where we describe the objects precisely as sets. However, there is an alternative perspective, which is just as useful, describing these objects via their *universal properties*, which roughly asks what makes each object special amongst all other such objects. The standard framework for a universal property is, “Object X is unique up to isomorphism among all objects having property P.”

Problem 3. Universal Property of Quotients.

Let R be a ring and $I \subseteq R$ an ideal. Let $\pi: R \rightarrow R/I$ be the quotient map. Suppose $\varphi: R \rightarrow S$ is a ring homomorphism such that $I \subseteq \ker(\varphi)$. Define a function

$$\tilde{\varphi}: R/I \rightarrow S \quad \text{by} \quad \tilde{\varphi}(r + I) = \varphi(r).$$

- (a) Prove that $\tilde{\varphi}$ is well-defined.
- (b) Prove that $\tilde{\varphi}$ is a ring homomorphism.
- (c) Prove that $\tilde{\varphi} \circ \pi = \varphi$.
- (d) Prove that $\tilde{\varphi}$ is the unique ring homomorphism $R/I \rightarrow S$ with the property that

$$\tilde{\varphi} \circ \pi = \varphi.$$

- (e) Explain in words what this says about the quotient ring R/I .

Solution.

- (a) Let $r + I = r' + I \in R/I$. Then, $r - r' \in I$.

$$\tilde{\varphi}(r + I) = \varphi(r) = \varphi(r + (r - r')) = \varphi(r') = \tilde{\varphi}(r' + I)$$

Hence, $\tilde{\varphi}$ is well-defined.

- (b) Let $\bar{a}, \bar{b} \in R/I$, we check if the three conditions hold.

Unital: Since φ is a ring homomorphism, we have

$$\tilde{\varphi}(1_{R/I}) = \varphi(1_R) = 1_S$$

Addition: Again, since φ is a ring homomorphism, we have

$$\tilde{\varphi}(\bar{a}) + \tilde{\varphi}(\bar{b}) = \varphi(a) + \varphi(b) = \varphi(a + b) = \tilde{\varphi}(\overline{a + b})$$

Multiplication: Again, φ is a ring homomorphism, we have

$$\tilde{\varphi}(\overline{\mathfrak{a}}) \cdot \tilde{\varphi}(\overline{\mathfrak{b}}) = \varphi(\mathfrak{a}) \cdot \varphi(\mathfrak{b}) = \varphi(\mathfrak{a} \cdot \mathfrak{b}) = \tilde{\varphi}(\overline{\mathfrak{a} \cdot \mathfrak{b}})$$

Hence, $\tilde{\varphi}$ is a ring homomorphism.

(c) Let $\mathfrak{a} \in \mathbf{R}$. Then,

$$\tilde{\varphi} \circ \pi(\mathfrak{a}) = \tilde{\varphi}(\overline{\mathfrak{a}}) = \varphi(\mathfrak{a})$$

Since $\mathfrak{a}$ was an arbitrary element in $\mathbf{R}$, this holds for all elements, and hence $\tilde{\varphi} \circ \pi = \varphi$.

(d) Assume there exists some other homomorphism $\sigma : \mathbf{R}/I \rightarrow \mathbf{S}$ such that $\sigma \circ \pi = \varphi$. Then for any $\overline{\mathfrak{a}} \in \mathbf{R}/I$,

$$\sigma(\overline{\mathfrak{a}}) = \sigma \circ \pi(\mathfrak{a}) = \varphi(\mathfrak{a}) = \tilde{\varphi}(\mathfrak{a})$$

(e) Every homomorphism φ such that $I \subseteq \ker(\varphi)$ can be written as $\tilde{\varphi} \circ \pi$ for some $\tilde{\varphi}$.

Problem 4. In this problem, we wish to consider the evaluation map:

$$\text{ev}_i: \mathbb{Z}[x] \rightarrow \mathbb{C}, \quad f(x) \mapsto f(i)$$

- (a) Prove that ev_i is a ring homomorphism.
- (b) Show that $\langle x^2 + 1 \rangle \subset \ker(\text{ev}_i)$.
- (c) Use Problem 3 to produce a ring homomorphism

$$\widetilde{\text{ev}}_i: \mathbb{Z}[x]/\langle x^2 + 1 \rangle \rightarrow \mathbb{C}.$$

- (d) What does $\widetilde{\text{ev}}_i$ do to the class of x in $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$?
- (e) Prove that $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$ is isomorphic to $\mathbb{Z}[i]$. Is $\langle x^2 + 1 \rangle$ prime and/or maximal?

Solution.

- (a) Let $f, g \in \mathbb{Z}[x]$, we check if the three conditions hold.

Unital: Let f_1 be the constant function 1. Then,

$$\text{ev}_i(f_1(x)) = f_1(i) = 1$$

Addition: Since $\mathbb{Z}[x]$ is a ring under pointwise addition and multiplication, we can write,

$$\text{ev}_i(f(x)) + \text{ev}_i(g(x)) = f(i) + g(i) = (f + g)(i) = \text{ev}_i((f + g)(x))$$

Multiplication: Since $\mathbb{Z}[x]$ is a ring under pointwise addition and multiplication, we can write,

$$\text{ev}_i(f(x)) \cdot \text{ev}_i(g(x)) = f(i) \cdot g(i) = (f \cdot g)(i) = \text{ev}_i((f \cdot g)(x))$$

Hence, ev_i is a ring homomorphism.

- (b) Note that $\text{ev}_i(x^2 + 1) = i^2 + 1 = -1 + 1 = 0$. For any $f(x)(x^2 + 1) \in \langle x^2 + 1 \rangle$ we have $\text{ev}_i(f(x)(x^2 + 1)) = f(i) \cdot 0 = 0 \in \ker(\text{ev}_i)$. Thus the ideal generated by $\langle x^2 + 1 \rangle$ under ev_i maps to 0 and hence is a subset of the kernel.

- (c) From Q3 we know that if $\langle x^2 + 1 \rangle \subseteq \ker(\text{ev}_i)$ there exists a unique homomorphism,

$$\widetilde{\text{ev}}_i: \mathbb{Z}[x]/\langle x^2 + 1 \rangle \rightarrow \mathbb{C}, \quad \widetilde{\text{ev}}_i(\overline{f(x)}) = f(i)$$

- (d) $\widetilde{\text{ev}}_i(\overline{x}) = x(i) = i$.

(e) Define a map $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}[i]$, $\varphi(f(x)) \rightarrow f(i)$. We will show that $\ker(\varphi) = \langle x^2 + 1 \rangle$.
 $\langle x^2 + 1 \rangle \subseteq \ker(\varphi)$

Note,

$$\varphi(x^2 + 1) = i^2 + 1 = 0 \in \ker(\varphi)$$

$$\ker(\varphi) \subseteq \langle x^2 + 1 \rangle$$

Let $\varphi(f(x)) \in \ker(\varphi)$. If $f(i) = 0$, then $x^2 + 1 \mid f(x)$ in $\mathbb{Z}[x]$ (since $x^2 + 1$ is the minimal polynomial of i over $\mathbb{Q}$ and hence is primitive over $\mathbb{Z}$), and we have,

$$f(x) \in \langle x^2 + 1 \rangle \implies \ker(\varphi) \subseteq \langle x^2 + 1 \rangle$$

Then by the first isomorphism theorem,

$$\mathbb{Z}[x]/\langle x^2 + 1 \rangle \cong \text{im}(\varphi) = \mathbb{Z}[i]$$

Since $\mathbb{Z}[x]$ is an integral domain but not a field, $\langle x^2 + 1 \rangle$ is prime but not maximal.

Problem 5. Universal Property of Polynomial Rings.

The goal of this exercise is to prove the following characterization of polynomial rings. Let $\varphi: R \rightarrow S$ be a ring homomorphism. For any element $s \in S$ there is a unique ring homomorphism $\psi: R[x] \rightarrow S$ such that $\psi(x) = s$ and the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & \nearrow \psi & \\ R[x] & & \end{array} .$$

- (a) Prove there is at most one ring homomorphism $\psi: R[x] \rightarrow S$ with the above properties.
- (b) Construct such a homomorphism explicitly by setting

$$\psi\left(\sum_{i=0}^n a_i x^i\right) = \sum_{i=0}^n \varphi(a_i) s^i.$$

- (c) Conclude that to give a ring homomorphism $R[x] \rightarrow S$ is equivalent to giving:
- a ring homomorphism $R \rightarrow S$, and
 - an element of S .

Solution.

- (a) Suppose there exists another homomorphism $\sigma: R[x] \rightarrow S$ such that $\sigma(x) = \psi(x) = s$. Also, note that since the diagram commutes $\psi|_R = \sigma|_R = \varphi$.

Then for any $\sum_{i=0}^n a_i x^i \in R[x]$ we have,

$$\begin{aligned} \sigma\left(\sum_{i=0}^n a_i x^i\right) &= \sigma(a_1) \cdot \sigma(x) + \cdots + \sigma(a_n) \cdot \sigma(x^n) \\ &= \varphi(a_1) \cdot \psi(x) + \cdots + \varphi(a_n) \cdot \psi(x^n) \\ &= \psi(a_1) \cdot \psi(x) + \cdots + \psi(a_n) \cdot \psi(x^n) \\ &= \psi\left(\sum_{i=0}^n a_i x^i\right) \end{aligned}$$

Hence, $\sigma = \psi$.

(b) Let $\psi : R[x] \rightarrow S$ such that

$$\psi(x) = s, \quad \psi(a) = \varphi(a) \quad \forall a \in R$$

(c) Hence for any ring homomorphism $\sigma : R[x] \rightarrow S$ we have

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & \nearrow \psi & \\ R[x] & & \end{array}$$

Problem 6. Evaluation maps.

Let R be a ring and let $a \in R$. Define the evaluation map

$$\text{ev}_a: R[x] \rightarrow R, \quad f(x) \mapsto f(a).$$

- (a) Prove that ev_a is a surjective ring homomorphism.
- (b) Prove that $\ker(\text{ev}_a) = \langle x - a \rangle$.
- (c) Use Problem 3 to show that ev_a factors through a ring homomorphism

$$R[x]/\langle x - a \rangle \rightarrow R.$$

- (d) Explain what this means if R is a field or R is an integral domain.
- (e) Give an example of a ring R and prime/maximal ideal $I \subset R[x]$ such that $I \neq \ker(\text{ev}_a)$ for all $a \in R$.

Solution.

- (a) For all $r \in R$ consider the constant functions r represented by $f_r \in R[x]$. Then,

$$\text{ev}_a(f_r(x)) = f_r(a) = r$$

- (b) $\ker(\text{ev}_a) \subseteq \langle x - a \rangle$.

Note that for any $p(x) \in \ker(\text{ev}_a)$ we have

$$p(x) = (x - a)q(x) + r(x)$$

for some $q(x), r(x) \in R[x]$. Evaluating at ev_a we have,

$$\text{ev}_a(p(x)) = (a - a)q(a) + r(a) = 0 \implies r(a) = 0$$

That is

$$p(x) = (x - a)q(x) \implies p(x) \in \langle x - a \rangle$$

$$\langle x - a \rangle \subseteq \ker(\text{ev}_a).$$

Let $p \in \langle x - a \rangle$. Since $\langle x - a \rangle$ irreducible we have $p = f(x)(x - a)$, for some $f(x) \in R[x]$,

$$\text{ev}_a(p) = \text{ev}_a(f(x)(x - a)) = f(a)(a - a) = 0 \in \ker(\text{ev}_a)$$

Hence, $\ker(\text{ev}_a) = \langle x - a \rangle$.

- (c) Since $\ker(\text{ev}_a) \supseteq \langle x - a \rangle$ from problem 3 we know we have an induced ring homomorphism $\widetilde{\text{ev}}_a$ such that

$$\begin{array}{ccc} \mathbf{R}[x] & \xrightarrow{\pi} & \mathbf{R}[x]/\langle x - a \rangle \\ \text{ev}_a \downarrow & \swarrow \widetilde{\text{ev}}_a & \\ \mathbf{R} & & \end{array}$$

- (d) If $\mathbf{R}$ is a domain this means $\langle x - a \rangle$ is prime and also that $\widetilde{\text{ev}}_a = \text{id}$.

If $\mathbf{R}$ is a field $\langle x - a \rangle$ is maximal and $\mathbf{R} \cong \mathbf{R}[x]/\langle x - a \rangle$. Also since all domains are fields we have $\widetilde{\text{ev}}_a = \text{id}$

- (e) $\mathbf{R} = \mathbb{C}, I = \langle 0 \rangle$.